Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Engineering & Technology
B.Tech. | Sem-6 | Winter-2025

Course Code	:	2101CS621	**Date**	:	11-11-2025
Course Name	:	Machine Learning	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	What is machine learning? Explain the difference between traditional programming and machine learning.	**4**
	(B)	Differentiate classification and regression.	**3**
		OR	
		Explain application of machine learning.	
	(C)	Explain in which situations machine learning is helpful.	**7**
		OR	
		Explain Bias-Variance Trade-Off with an example.	
Q.2	**(A)**	Explain Chi-Square Test with an example.	**4**
	(B)	Differentiate Linear Regression vs Logistic Regression	**3**
		OR	
		Explain Ordinary Least Squares with an example.	
	(C)	What is Logistic Regression? Explain the mathematical formulation and write a Python code to apply Logistic Regression on the Iris dataset.	**7**
		OR	
		What is PCA? Write a python code to convert a dataset with 12 features into 2 principal components using PCA and visualize the transformed dataset. (generate data randomly).	
Q.3	**(A)**	Explain accuracy, precision, recall and F1 with an example.	**4**
	(B)	Explain bootstrap with an example.	**3**
		OR	
		Explain cross validation with an example.	
	(C)	What is SVM? Explain kernel of SVM with an example.	**7**
		OR	
		Explain random forest algorithm with an example.	

Q.4	**(A)**	Write a Python program to implement k-means clustering with randomly generated data points.	**4**
	(B)	Explain types of outliers.	**3**
		OR	
		Explain a dendrogram with an example.	
	(C)	Explain Agglomerative Clustering with an example.	**7**
		OR	
		Differentiate k-means and K-medoids, and explain the working of k-medoids algorithm with an example.	
Q.5	**(A)**	Explain backpropagation with an example.	**4**
	(B)	Explain Softmax activation function.	**3**
		OR	
		Explain Leaky ReLU activation function.	
	(C)	Explain hyperparameters with an example.	**7**
		OR	
		Explain CNN with an example.	

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